



Maintenance Schedule

Diesel engine

16V2000G26S

Application group 3B 3F 3G

MS50209/03E

Engine model	kW/cyl.	Application group
16V2000G26S	62,4 kW/cyl.	3B, Continuous operation, variable load, ICXN 3F, Mains backup mode, unlimited, ICXN 3G, Prime power, time-limited, ICXN

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Allocation matrix application group 3B, 3F, 3G

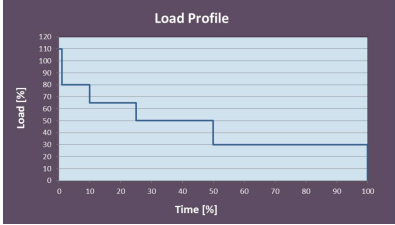
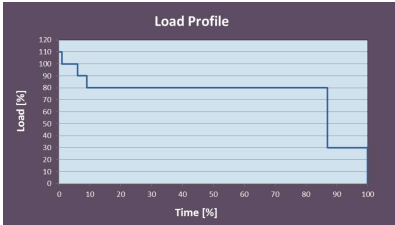
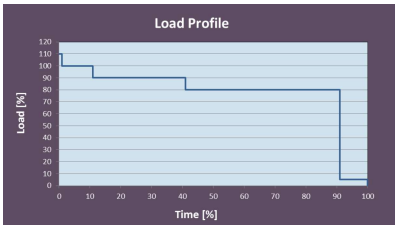
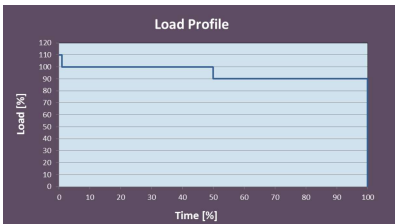
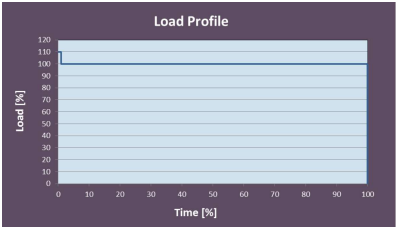
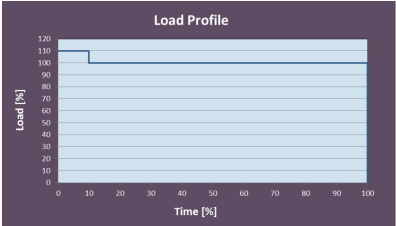
Load profile		16V2000G26S						
Load %	Time %	Load factor %	Load indicator %	16V2000G26S				
110 ₍₁₀₀₋₁₁₀₎	1	0 to 46	≤ 5	24,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	—							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	9							
65 ₍₅₀₋₆₅₎	15							
50 ₍₃₀₋₅₀₎	25							
30 ₍₅₋₃₀₎	50							
Idle Speed ₍₀₋₅₎	—							
110 ₍₁₀₀₋₁₁₀₎	1	47 to 75	≤ 18	24,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	3							
80 ₍₆₅₋₈₀₎	78							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	13							
Idle Speed ₍₀₋₅₎	—							
110 ₍₁₀₀₋₁₁₀₎	1	76 to 80	≤ 32	22,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	10							
90 ₍₈₀₋₉₀₎	30							
80 ₍₆₅₋₈₀₎	50							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle Speed ₍₀₋₅₎	9							
110 ₍₁₀₀₋₁₁₀₎	1	81 to 95	≤ 70	18,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	49							
90 ₍₈₀₋₉₀₎	50							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle Speed ₍₀₋₅₎	—							

Table 2: Allocation matrix application group 3B, 3F, 3G

TIM-ID: 0000064561 - 002

16V2000G26S								
Load profile		Load factor %	Load indicator %	16V2000G26S				
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	1	96 to 100	≤ 102	14,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	99							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle Speed ₍₀₋₅₎	—	100 to 101	≤ 116	14,000	—	—	—	
110 ₍₁₀₀₋₁₁₀₎	10*							
100 ₍₉₀₋₁₀₀₎	90							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle Speed ₍₀₋₅₎	—							

* no continuous use by 110% power

Table 3: Allocation matrix application group 3B, 3F, 3G

1.4 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0010	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
0020	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.	WM00097	1000	0	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.5 Maintenance Schedule Matrix 24000 HRS

0 to 14500 Operating hours

Task	Limit	Operating hours [HRS]																													
		Daily	500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500
WM00296	1 MON																														
WM00067	2 YR																														
WM00297	9 YR																														
WM00172	18 YR																														
WM00285	-	X																													
WM00286	-	X																													
WM00287	-	X																													
WM00288	-	X																													
WM00289	-	X																													
WM00086	2 YR		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00050	1 YR			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
WM00136	1 YR			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
WM00094	18 YR			X		X				X				X				X				X				X		X		X	
WM00170	2 YR									X								X								X					
WM00076	3 YR									X								X									X				
WM00106	6 YR													X												X					
WM00110	9 YR																	X													
WM00090	18 YR																	X													
WM00154	18 YR																	X													
WM00186	18 YR																	X													
WM00126	5 YR																										X				
WM00005	6 YR																										X				
WM00006	6 YR																										X				
WM00007	6 YR																										X				
WM00008	6 YR																										X				
WM00003	18 YR																										X				
WM00004	18 YR																										X				
WM00054	18 YR																										X				
WM00056	18 YR																										X				
WM00059	18 YR																										X				
WM00060	18 YR																										X				
WM00097	18 YR																										X				
WM00101	18 YR																										X				
WM00109	18 YR																										X				
WM00117	18 YR																										X				
WM00133	18 YR																										X				
WM00164	18 YR																										X				
WM00165	18 YR																										X				
WM00166	18 YR																										X				
WM00011	18 YR																														
WM00051	18 YR																														
WM00071	18 YR																														

MON = Months

YR = Years

HRS = Hours

TM4D: 000000000 - 000

Task	Limit	Operating hours [HRS]																													
		Daily	500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500
WM00091	18 YR																														
WM00001	18 YR																														
WM00002	18 YR																														
WM00088	18 YR																														
WM00099	18 YR																														
WM00100	18 YR																														
WM00102	18 YR																														
WM00103	18 YR																														
WM00104	18 YR																														
WM00108	18 YR																														
WM00111	18 YR																														
WM00127	18 YR																														
WM00162	18 YR																														
WM00222	1 YR																														
WM00148	3 YR																														
WM00040	18 YR																														
WM00228	6 MON		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00188	1 YR			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
WM00189	1 YR			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
WM00190	5 YR																									X					
WM00020	18 YR																														
WM00215	1 MON																														
WM00216	1 MON																														
WM00214	2 YR		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
MON = Months																															
YR = Years																															
HRS = Hours																															

15000 to 24000 Operating hours

Task	Limit	Operating hours [HRS]																													
		15000	15500	16000	16500	17000	17500	18000	18500	19000	19500	20000	20500	21000	21500	22000	22500	23000	23500	24000											
WM00296	1 MON																														
WM00067	2 YR																														
WM00297	9 YR																														
WM00172	18 YR																														
WM00285	-																														
WM00286	-																														
WM00287	-																														
WM00288	-																														
WM00289	-																														
WM00086	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X											
WM00050	1 YR	X		X		X		X		X		X		X		X		X		X											
WM00136	1 YR	X		X		X		X		X		X		X		X		X		X											
WM00094	18 YR			X				X				X				X				X											
WM00170	2 YR			X								X								X											
MON = Months																															
YR = Years																															
HRS = Hours																															